MATTHEW WU

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, College of Engineering — Atlanta, Georgia
Bachelor of Science in Computer Engineering — *June 2024 - May 2028*
GPA: 3.75/4.0

- Concentration: Distributed Systems & Software Design and Computing Hardware & Emerging Architectures
- Scholarships: Zell Miller (100% Tuition), Georgia Tech Institutional Scholarship
- Relevant Coursework: Circuit Analysis, Physical Foundations of Computer Engineering, Digital Systems Design, Computer Architecture, Data Structures and Algorithms, Operating Systems, Systems Administration, Machine Learning

PROFESSIONAL EXPERIENCE

GE AEROSPACE – Engines Engineering — Evendale, Ohio
Engines Engineering Co-op (Computer/Software) — *January 2027 – May 2027*

EMBEDDED AND AUTONOMOUS SYSTEMS LABORATORY – Georgia Institute of Technology — Atlanta, Georgia
Undergraduate Research Assistant — *May 2026 – August 2026*

- Developed an LLM-driven pipeline generating SystemVerilog test benches from Intel FPGA vendor docs, cutting development time 38% (Python).
- Built a PDF/HTML-to-module-spec extraction pipeline enabling 85% first-pass compile success across 20 Intel FPGA IP cores.
- Automated simulation/waveform validation in QuestaSim, achieving 78% functional coverage against reference test benches.
- Reduced hallucinated port/signal errors from 32% to 7% by refining LLM prompts using Python-based compile-log analysis and automated failure logging.

SOFTWARE SYSTEMS ENGINEER — HUA FOODS (Postpartum Nutrition Service) — Atlanta, Georgia
Developer — *August 2025 - Present*

- Developed a centralized software platform that automated customer intake, meal selection, scheduling, and delivery coordination, reducing customer onboarding time by 83% and supporting business growth from approximately $85K to $130K in annual revenue (JavaScript).
- Developed an ESP32 automated packaging system that reduced package time from 2 minutes to 15 seconds, reducing significant manual labor (C++).

FREELANCE SOFTWARE DEVELOPER — Remote
Developer — *August 2021 – August 2024*

- Built and deployed React websites for local businesses without an online presence, generating over $5K in freelance revenue.
- Integrated Firebase for contact forms and dynamic content, while debugging and refactoring code to improve reliability.

PROJECTS

AI TRADING ASSISTANT – Personal Project — Atlanta, Georgia
Developer — *June 2026 - Present*

- Built LLM trading agent on Robinhood Agentic API via MCP, shaped by 200+ Reddit posts (10 communities); 15+ Discord users report better discipline (Python).
- Built backtester turning plain-English rules into code on 6 mo. of 5-min data; cut validation 2 hrs→5 min, rejecting 3/5 strategies under 70% win rate.
- Added Pine Script v6 and risk guardrails (order review, account isolation, trade limits); win rate 45%→78% over 1,000+ trades, weekly return +150% in 20 wks.

AUTONOMOUS MONITOR TRACKING SYSTEM – Personal Project — Atlanta, Georgia
Developer — *March 2026 - April 2026*

- Designed a real-time human-tracking system to auto-orient a monitor toward the user across activities like workouts, calls, and studying, using OpenCV and Kalman filtering to achieve 95% target lock accuracy with under 75ms end-to-end latency (Python).
- Developed and tuned ESP32 firmware for a 200Hz closed-loop PID system, balancing responsiveness and stability to track targets up to 3m away (C++).
- Integrated and debugged the full software, firmware, and hardware stack, validating reliable end-to-end system performance across 20+ test sessions.

ADAPTIVE WEARABLE GAMING CONTROLLER | TECHHACK Valley Hackathon — Atlanta, Georgia
Developer | Team of 4 — *March 2026*

- Designed an adaptive gaming controller for users with limited mobility, translating wrist gestures into HID inputs to improve gaming accessibility.
- Developed ESP32 firmware implementing IMU sensor fusion, orientation estimation, and gesture recognition at 300Hz, achieving 98% gesture accuracy.
- Debugged and optimized firmware/input pipeline to achieve 12ms response latency across 6 customizable gesture mappings within a 36-hour hackathon.

ARDUINO TASK WATCH – Class Project — Atlanta, Georgia
Developer — *November 2024 - January 2025*

- Engineered a wearable ESP32-based task verification device using GPS geofencing to confirm scheduled tasks within a 15 m radius.
- Developed embedded firmware coordinating 5 hardware peripherals through an 8-state finite-state machine, processing user inputs at 100Hz (C++).
- Repurposed and soldered a commercial electrical stimulation module with 3 adaptive intensity levels based on task completion delay.

LEADERSHIP

GITMAD (Georgia Institute of Technology Mastering Android Development) — Atlanta, Georgia
Engineering Lead — *December 2025 - Present*

- Led development of student Android application projects, guiding architecture decisions and feature implementation.
- Mentored team members in debugging, code structure, and collaborative software development practices.

SEECHINA INTERNATIONAL ENGINEERING EXCHANGE — China
Communications & Data Lead — *May 2025 - July 2025*

- Led technical discussions with professors and directors at top Chinese universities (Tsinghua, Peking, Zhejiang, Fudan) and top Chinese tech companies (Tencent, BYD, Xiaomi, DJI, XAG) on China's advancements in AI and industrial production.

FOREIGN LANGUAGE EDUCATION PROGRAM — Taiwan
Student Body President / Lead Foreign English Teacher — *May 2023 - August 2023*

- Recognized by the Taiwan Governor and featured in national media for leading initiatives, including 60+ workshops that trained 750+ foreign educators.

SKILLS

Certifications	AWS Certified Solutions Architect – Associate; AWS Certified AI Practitioner
Languages	English – native; Mandarin – native; Spanish – intermediate
Artificial Intelligence/ML	PyTorch, OpenCV
Programming	C++, Python, Java, JavaScript, MATLAB, System Verilog
Embedded Systems	ESP32, Raspberry Pi, PlatformIO, VHDL, RISC-V Assembly
Development Tools	Git, Linux, Bash, Quartus, Multisim, QuestaSim, Vivado, GitHub, Excel